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Activity-7

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Correlation Detective Case file (D-C14)

Q1) The matrix

- No of observations: n = 15
- No of variables: 7
- N of distinct variable pairs: $\frac{7(7-1)}{2} = 21$

R-code

```
set.seed(472380)
n <- 15
pressure <- round(rnorm(n, 59.2, 11.9), 1)
group_sessions <- round(rnorm(n, 12, 13.0), 0)
print_jobs <- round(237 + 595 * (0.11 * (group_sessions - 12)/3.0 + 0.99)
group_sessions[6] <- 22
print_jobs[6] <- 42)
library(ggplot2)
geom_title(color = "white"
labs(title = "correlation Detective: correlogram", x = "", y = "", fill =
"correlation")
```

Q2) The strongest clues

```
C2 <- C
diag(c2) <- NA
upper <- which(upper.tri(c2), arr.ind = TRUE)
pair_values <- data.frame(
  V1 = rownames(c2)[upper[, 1],
  V2 = c2[upper],
```

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```
positive <- pair_values[which.max(pair_values^2)]

print(positive)

classify <- function(r) {

  if(abs(r) < 0.3) "weak"

  else if (abs(r) <= 0.7) "Moderate"

  else "strong"

}

cat("Positive relationship:", classify(positive^2), "\n")
```

Strongest positive clue: pressure and coffee

Q3) The deceptive Zero

```
plot(study_complaints ~ room_dev, data = case,

  main = "study complaints vs Room Temperature Deviation",

  xlab = "Room Temperature Deviation",

  ylab = "study complaints",

  pch = 19, col = "blue")

x_grid <- seq(min(cases$room_dev), max(cases$room_dev),

              length.out = 200)

lines(x_grid, predict(quadratic_fit, new data = data.frame(room_dev =

              x_grid)),

      col = "darkgreen", lwd = 2)

cat("Pearson correlation r =", round(r_linear, 4), "\n")
```

Pearson's correlation r is expected to be close to zero, while the quadratic model should show a substantially stronger fit.

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Q4) The outlier alibi

```
plot(print_jobs ~ groups_sessions, data=case,
     main = "Print jobs vs Group sessions",
     xlab = "Group sessions",
     ylab = "print jobs",
     pch = 19, col = "blue")
text(case$group_sessions, case$print_jobs,
     labels = 1:nrow(case), pos=3, cex=0.8)

cat("correlation with all rows:", round(r_all, 4), "\n")
cat("suspicious row = 6\n")
```

Q5) The prime suspect

```
r_coffee_late <- cor(case$coffee, case$late_entries)
r_pressure_coffee <- cor(case$pressure, case$coffee)
res_coffee <- resid(lm(coffee ~ pressure, data=case))
res_late <- resid(lm(late_entries ~ pressure, data=case))
partial_r <- cor(res_coffee, res_late)
cat("partial correlation controlling for pressure:",
    round(partial_r, 4), "\n")
```

Two-sentence verdict:

Coffee sales do not necessarily cause late-night entries, because exam pressure may influence both variables & create a misleading correlation.

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